Kramer Annulus Stokes Benchmark Using Underworld3

Thyagarajulu Gollapalli[1] and Underworld Team[1]

[1]Research School of Earth Sciences, Australian National University, Canberra, ACT, Australia

Correspondence: Thyagarajulu Gollapalli (gthyagi@gmail.com)

Abstract. We reproduce the annulus benchmark of Kramer et al. (2021) using Underworld3 (UW3). The benchmark solves incompressible, isoviscous Stokes flow in a two-dimensional cylindrical shell with analytical forcing and analytical velocity and pressure solutions. It tests curved boundaries, pressure normalisation, free-slip and zero-slip velocity conditions, and both smooth and interface-localised density forcing. The UW3 calculations use $P_2 \times P_1$ Taylor–Hood finite elements on refined annulus meshes with nominal cell sizes $h = 1/8, \ldots, 1/256$. Errors are measured using relative volume L_2 norms for velocity and pressure. Smooth forcing cases recover approximately second-order pressure convergence and systematic second-order or better velocity convergence over the tested refinement range. Delta-function forcing cases show reduced but regular convergence, with velocity rates near $O(h^{1.5})$ and pressure rates near $O(h^{0.5})$, consistent with the reduced regularity of interface-localised buoyancy forcing. These results verify the UW3 annulus Stokes implementation for curved-shell geometry and for both boundary-condition families considered in the Kramer benchmark.

1 Introduction

Analytical benchmarks are central to verification of geodynamic Stokes-flow codes because they separate discretisation and implementation errors from uncertainty in the reference solution. Annulus benchmarks are particularly useful for mantle-convection software: they include curved boundaries and polar coordinate forcing while remaining cheaper and easier to inspect than full spherical-shell calculations.

The analytical solutions of Kramer et al. (2021) provide a family of mantle-flow benchmarks in cylindrical and spherical shells. In the cylindrical annulus, the solutions include free-slip and zero-slip boundary conditions, smooth volumetric forcing, and delta-function forcing on an internal circular interface. This combination exercises several aspects of a finite-element Stokes implementation: velocity–pressure coupling, pressure gauge removal, curved-domain integration, normal and tangential boundary constraints, and the treatment of fields with limited regularity.

Here we summarise the UW3 reproduction of the Kramer annulus benchmark. The article is based on the corresponding UW3 benchmark report and uses only the existing generated figures and tables. The goal is to document the mathematical setup, numerical choices, error definitions, and observed convergence behaviour in a concise Copernicus-style manuscript.

2 Governing Equations and Benchmark Solution

The benchmark solves the steady incompressible Stokes equations in a two-dimensional annulus $\Omega = \{(r,\theta) : R_- \le r \le R_+\}$:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{1b}$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \qquad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right), \tag{2}$$

with constant viscosity η.

Following Kramer et al. (2021), the annulus radii are

$$R_- = 1.22, \qquad R_+ = 2.22.$$

For the interface-localised cases, the internal density interface is located at

$$R_{\mathrm{int}} = 2.0.$$

The analytical forcing is separated into azimuthal and radial parts. Smooth cases use a density perturbation proportional to

$$\left(\frac{r}{R_+}\right)^k \cos(n\theta), \tag{3}$$

where k controls the radial structure and n controls the azimuthal wavenumber. Delta-function cases use an interface forcing proportional to

$$\delta(r - R_{\mathrm{int}})\cos(n\theta). \tag{4}$$

The full analytical velocity and pressure fields are those derived by Kramer et al. (2021); the UW3 implementation evaluates these closed-form solutions to define the forcing, impose the benchmark boundary conditions, and measure numerical error.

3 Numerical Setup

The UW3 calculations use $P_2 \times P_1$ Taylor–Hood elements, with quadratic velocity and continuous linear pressure. This mixed pair is a standard choice for incompressible Stokes flow and is expected to provide robust pressure behaviour for smooth solutions (Girault and Raviart, 1986; Boffi et al., 2013). The pressure nullspace is removed before error evaluation because incompressible Stokes flow determines pressure only up to an additive constant.

The verification suite contains four benchmark groups (Table 1). Cases 1 and 2 use free-slip boundaries, while Cases 3 and 4 use zero-slip boundaries. Cases 1 and 3 use delta-function forcing on the internal interface; Cases 2 and 4 use smooth volumetric forcing. The tested nominal mesh sizes are $h = 1/8, \ldots, 1/256$.

Table 1. Kramer annulus benchmark cases used in the UW3 verification runs.

Case	Forcing	Boundary condition	Parameters	Resolutions
1	Delta function	Free slip	$n = 2, 8, 32$	$1/8$–$1/256$
2	Smooth	Free slip	$n = 2, 8, 32;\ k = 2, 8$	$1/8$–$1/256$
3	Delta function	Zero slip	$n = 2, 8, 32$	$1/8$–$1/256$
4	Smooth	Zero slip	$n = 2, 8, 32;\ k = 2, 8$	$1/8$–$1/256$

Figure 1 shows representative analytical density perturbations. Increasing n increases the angular complexity of the benchmark solution. For smooth forcing, increasing k changes the radial weighting of the density field. For delta-function forcing, the density is concentrated on the internal interface and therefore reduces solution regularity relative to the smooth cases.

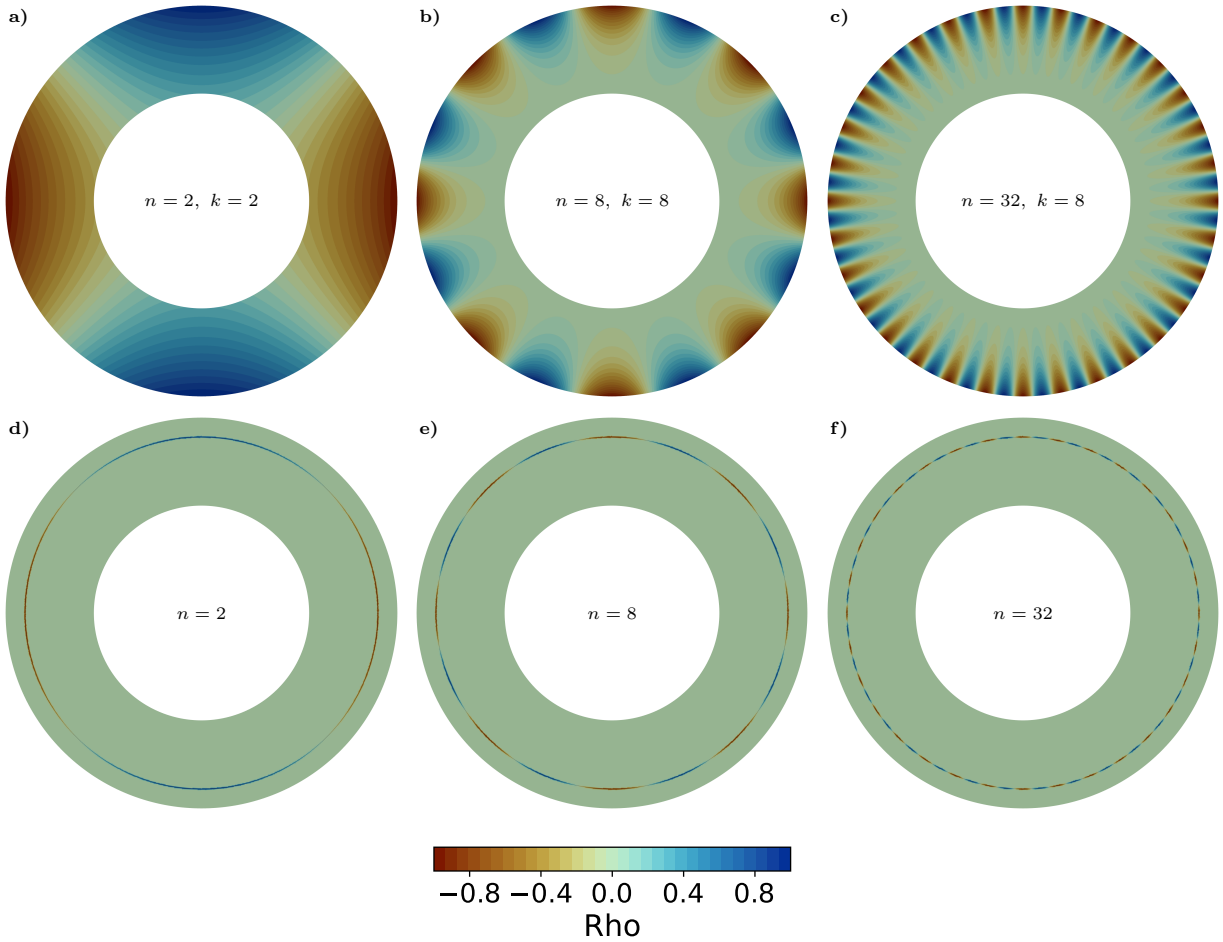

Figure 1. Analytical density perturbations for the Kramer annulus benchmark. The upper row shows smooth forcing and the lower row shows delta-function forcing. Columns correspond to increasing azimuthal complexity.

4 Error Metrics

Velocity and pressure errors are reported as relative volume L_2 norms. For a numerical field q_h and analytical reference field q^*, the relative error is

$$E_{L_2}^{\mathrm{rel}}(q) = \left(\frac{\int_\Omega |q_h - q^*|^2 \, \mathrm{d}\Omega}{\int_\Omega |q^*|^2 \, \mathrm{d}\Omega} \right)^{1/2}. \tag{5}$$

Thus the table and figure labels use $E_{L_2}^{\mathrm{rel}}(\mathbf{u})$ for velocity and $E_{L_2}^{\mathrm{rel}}(p)$ for pressure. For two consecutive refinements with characteristic mesh sizes h_1 and h_2, the observed convergence rate is

$$\alpha = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}. \tag{6}$$

With uniform halving of the nominal mesh size, this reduces to $\alpha = \log_2(E_{h_1}/E_{h_2})$.

5 Results

The convergence results are shown in Fig. 2. For the smooth forcing cases, pressure errors converge close to $O(h^2)$ for both free-slip and zero-slip boundary conditions. Velocity errors converge systematically and are approximately second order for $n = 2$ and $n = 8$. The high-wavenumber $n = 32$ smooth cases show larger error constants and stronger pre-asymptotic behaviour, but the rates decrease toward the expected refined-mesh trend as the solution becomes better resolved.

The delta-function forcing cases exhibit lower rates. Velocity convergence approaches $O(h^{1.5})$, while pressure convergence approaches $O(h^{0.5})$ for both free-slip and zero-slip boundaries. This behaviour is expected for the Kramer benchmark because the Green's-function solution associated with interface-localised forcing is less regular than the smooth volumetric-forcing solution (Kramer et al., 2021). The reduced pressure rate is therefore interpreted as a property of the benchmark solution regularity rather than as a failure of the Stokes solver.

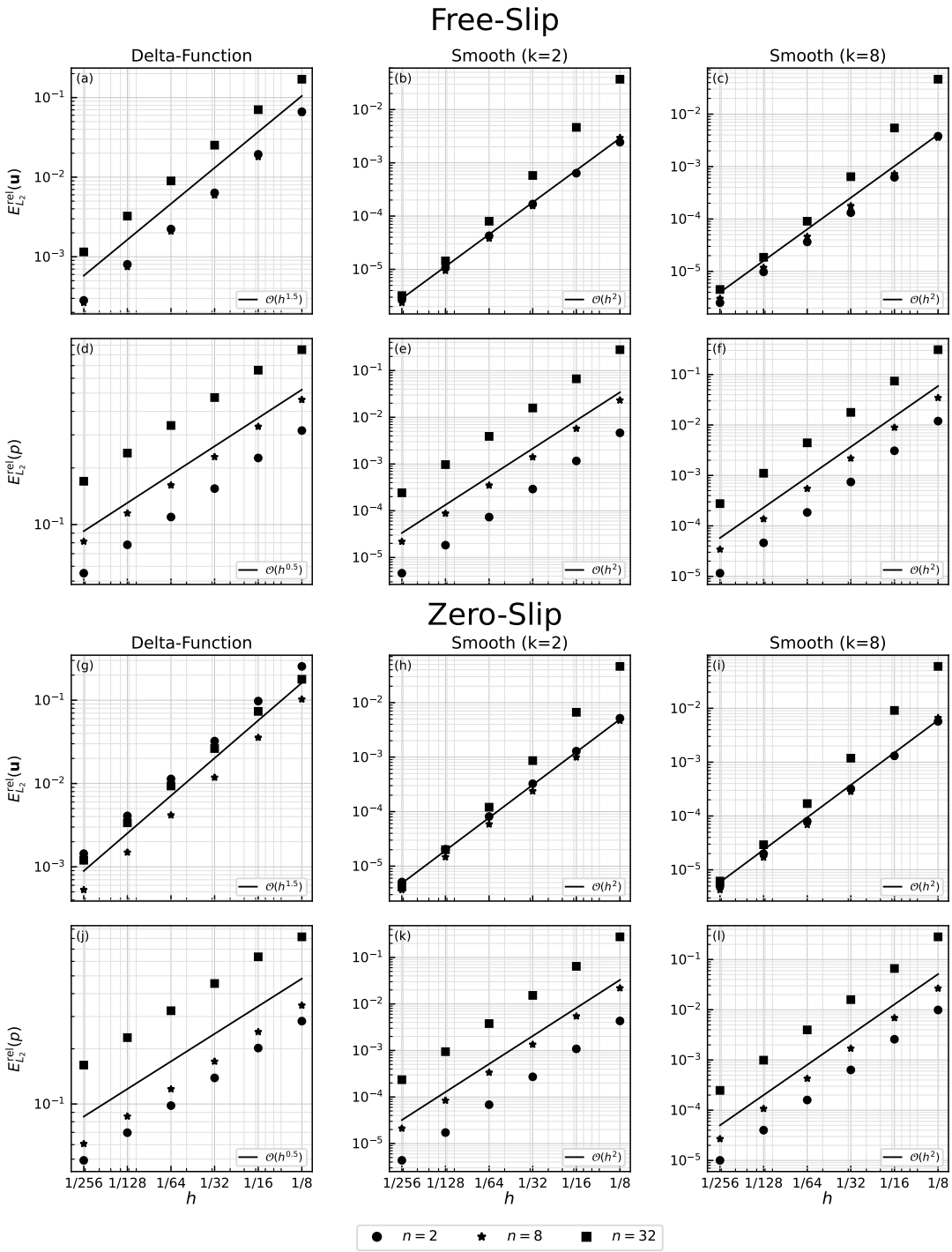

Figure 2. Velocity and pressure convergence for the four Kramer annulus cases. Columns show delta-function forcing and smooth forcing with $k = 2$ and $k = 8$. The upper block uses free-slip boundaries and the lower block uses zero-slip boundaries.

The numerical errors and pairwise convergence rates are provided in Tables 2–7. Each table groups three azimuthal wavenumbers and reports mesh size, velocity error, velocity rate, pressure error, and pressure rate.

Table 2. Free-slip, delta-function forcing.

h	$n=2$				$n=8$				$n=32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	6.636e-02	–	3.161e-01	–	6.455e-02	–	4.610e-01	–	1.700e-01	–	8.511e-01	–
$1/16$	1.936e-02	1.78	2.260e-01	0.48	1.798e-02	1.84	3.318e-01	0.47	7.045e-02	1.27	6.624e-01	0.36
$1/32$	6.346e-03	1.61	1.553e-01	0.54	5.914e-03	1.60	2.290e-01	0.54	2.523e-02	1.48	4.736e-01	0.48
$1/64$	2.230e-03	1.51	1.097e-01	0.50	2.084e-03	1.50	1.617e-01	0.50	8.968e-03	1.49	3.364e-01	0.49
$1/128$	8.026e-04	1.47	7.804e-02	0.49	7.460e-04	1.48	1.149e-01	0.49	3.242e-03	1.47	2.400e-01	0.49
$1/256$	2.828e-04	1.50	5.512e-02	0.50	2.643e-04	1.50	8.132e-02	0.50	1.150e-03	1.50	1.699e-01	0.50

Table 3. Free-slip, smooth forcing ($k=2$).

h	$n=2$				$n=8$				$n=32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	2.441e-03	–	4.639e-03	–	2.935e-03	–	2.313e-02	–	3.710e-02	–	2.796e-01	–
$1/16$	6.377e-04	1.94	1.164e-03	2.00	6.481e-04	2.18	5.728e-03	2.01	4.628e-03	3.00	6.660e-02	2.07
$1/32$	1.680e-04	1.92	2.911e-04	2.00	1.547e-04	2.07	1.401e-03	2.03	5.784e-04	3.00	1.580e-02	2.08
$1/64$	4.260e-05	1.98	7.312e-05	1.99	3.792e-05	2.03	3.487e-04	2.01	7.999e-05	2.85	3.904e-03	2.02
$1/128$	1.076e-05	1.99	1.836e-05	1.99	9.436e-06	2.01	8.722e-05	2.00	1.438e-05	2.48	9.718e-04	2.01
$1/256$	2.704e-06	1.99	4.611e-06	1.99	2.359e-06	2.00	2.189e-05	1.99	3.209e-06	2.16	2.425e-04	2.00

Table 4. Free-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	3.794e-03	–	1.195e-02	–	3.584e-03	–	3.454e-02	–	4.638e-02	–	3.106e-01	–
$1/16$	6.230e-04	2.61	3.080e-03	1.96	7.176e-04	2.32	8.903e-03	1.96	5.447e-03	3.09	7.427e-02	2.06
$1/32$	1.320e-04	2.24	7.416e-04	2.05	1.772e-04	2.02	2.180e-03	2.03	6.411e-04	3.09	1.777e-02	2.06
$1/64$	3.650e-05	1.85	1.848e-04	2.00	4.601e-05	1.95	5.461e-04	2.00	9.064e-05	2.82	4.427e-03	2.00
$1/128$	9.792e-06	1.90	4.617e-05	2.00	1.177e-05	1.97	1.367e-04	2.00	1.858e-05	2.29	1.105e-03	2.00
$1/256$	2.524e-06	1.96	1.154e-05	2.00	3.023e-06	1.96	3.424e-05	2.00	4.520e-06	2.04	2.761e-04	2.00

Table 5. Zero-slip, delta-function forcing.

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	2.545e-01	–	2.832e-01	–	1.025e-01	–	3.449e-01	–	1.783e-01	–	8.167e-01	–
$1/16$	9.755e-02	1.38	2.019e-01	0.49	3.555e-02	1.53	2.473e-01	0.48	7.344e-02	1.28	6.355e-01	0.36
$1/32$	3.228e-02	1.60	1.388e-01	0.54	1.181e-02	1.59	1.707e-01	0.54	2.629e-02	1.48	4.544e-01	0.48
$1/64$	1.134e-02	1.51	9.797e-02	0.50	4.163e-03	1.50	1.205e-01	0.50	9.343e-03	1.49	3.228e-01	0.49
$1/128$	4.083e-03	1.47	6.972e-02	0.49	1.490e-03	1.48	8.563e-02	0.49	3.378e-03	1.47	2.302e-01	0.49
$1/256$	1.439e-03	1.50	4.924e-02	0.50	5.281e-04	1.50	6.061e-02	0.50	1.198e-03	1.50	1.630e-01	0.50

Table 6. Zero-slip, smooth forcing ($k = 2$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	5.169e-03	–	4.309e-03	–	4.692e-03	–	2.188e-02	–	4.621e-02	–	2.766e-01	–
$1/16$	1.287e-03	2.01	1.083e-03	1.99	9.929e-04	2.24	5.433e-03	2.01	6.648e-03	2.80	6.437e-02	2.10
$1/32$	3.253e-04	1.98	2.715e-04	2.00	2.375e-04	2.06	1.343e-03	2.02	8.624e-04	2.95	1.527e-02	2.08
$1/64$	8.137e-05	2.00	6.831e-05	1.99	5.873e-05	2.02	3.358e-04	2.00	1.206e-04	2.84	3.784e-03	2.01
$1/128$	2.041e-05	2.00	1.721e-05	1.99	1.467e-05	2.00	8.419e-05	2.00	2.004e-05	2.59	9.432e-04	2.00
$1/256$	5.103e-06	2.00	4.351e-06	1.98	3.662e-06	2.00	2.111e-05	2.00	4.126e-06	2.28	2.354e-04	2.00

Table 7. Zero-slip, smooth forcing ($k = 8$).

h	$n = 2$				$n = 8$				$n = 32$			
	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate	$E^{\mathrm{rel}}_{L_2}(\mathbf{u})$	rate	$E^{\mathrm{rel}}_{L_2}(p)$	rate
$1/8$	5.720e-03	–	9.905e-03	–	6.657e-03	–	2.663e-02	–	6.009e-02	–	2.829e-01	–
$1/16$	1.311e-03	2.13	2.585e-03	1.94	1.283e-03	2.38	6.863e-03	1.96	9.136e-03	2.72	6.628e-02	2.09
$1/32$	3.202e-04	2.03	6.338e-04	2.03	2.837e-04	2.18	1.696e-03	2.02	1.185e-03	2.95	1.590e-02	2.06
$1/64$	7.962e-05	2.01	1.596e-04	1.99	6.861e-05	2.05	4.264e-04	1.99	1.699e-04	2.80	3.969e-03	2.00
$1/128$	1.990e-05	2.00	4.003e-05	2.00	1.700e-05	2.01	1.069e-04	2.00	2.923e-05	2.54	9.910e-04	2.00
$1/256$	4.970e-06	2.00	1.003e-05	2.00	4.234e-06	2.01	2.680e-05	2.00	6.181e-06	2.24	2.476e-04	2.00

6 Discussion

The convergence behaviour separates primarily by forcing type. Smooth forcing preserves the regularity needed for second-order pressure convergence with the $P_2 \times P_1$ discretisation. Velocity convergence is also regular, although the observed rates are influenced by the curved geometry, boundary-condition enforcement, and the number of elements per azimuthal wavelength. These effects are most visible for $n = 32$, where the coarsest meshes under-resolve the angular structure.

The delta-function cases are more demanding because the forcing is concentrated on an internal circular interface. The slower convergence rates are consistent with the reference Kramer solutions and with the expectation that pressure is especially sensitive to reduced regularity in mixed Stokes discretisations. Importantly, the free-slip and zero-slip variants show the same qualitative rates, indicating that the UW3 boundary-condition implementations behave consistently across both benchmark families.

The benchmark is therefore a useful intermediate verification problem between Cartesian manufactured solutions and full spherical-shell mantle-convection calculations. It tests curved-domain Stokes assembly, pressure normalisation, boundary conditions, and both smooth and non-smooth buoyancy forcing while remaining compact enough for routine convergence testing.

7 Conclusions

We have presented a Copernicus-style summary of the UW3 Kramer annulus Stokes benchmark. The main findings are:

1. UW3 reproduces the analytical annulus solutions of Kramer et al. (2021) for both free-slip and zero-slip velocity boundary conditions.

2. Smooth forcing cases show approximately second-order pressure convergence and systematic second-order or better velocity convergence over the tested mesh sequence.

3. Delta-function forcing cases show the expected reduced convergence, approximately $O(h^{1.5})$ in velocity and $O(h^{0.5})$ in pressure.

4. The observed distinction between smooth and interface-localised forcing reflects benchmark solution regularity and provides a useful test of finite element behaviour in curved geodynamic domains.

These results verify the UW3 implementation for cylindrical-shell Stokes flow with analytical Kramer forcing and provide a compact reference for future annulus and spherical-shell benchmark comparisons.

References

Boffi, D., Brezzi, F., and Fortin, M.: Mixed Finite Element Methods and Applications, Springer Berlin Heidelberg, ISBN 9783642365195, https://doi.org/10.1007/978-3-642-36519-5, 2013.

Girault, V. and Raviart, P.-A.: Finite Element Methods for Navier–Stokes Equations, Springer Berlin Heidelberg, ISBN 9783642616235, https://doi.org/10.1007/978-3-642-61623-5, 1986.

Kramer, S. C., Davies, D. R., and Wilson, C. R.: Analytical solutions for mantle flow in cylindrical and spherical shells, Geoscientific Model Development, 14, 1899–1919, https://doi.org/10.5194/gmd-14-1899-2021, 2021.